



Broadband tunable electro-optic switch/power divider as potential building blocks in integrated lithium niobate photonics

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Abstract: We demonstrate an electro-optic (EO) switch or in general, an EO controllable power divider based on a periodically poled lithium niobate (PPLN) polarization mode converter (PMC) and a five-waveguide adiabatic coupler integrated on a Ti:LN photonic circuit chip. In this integrated photonic circuit (IPC) device, the PPLN works as an EO controllable polarization rotator (and therefore a PMC), while the adiabatic coupler functions as a broadband polarization beam splitter (PBS). The 1-cm long PPLN EO PMC of the IPC device is characterized to have a half-wave (or switching) voltage of $V_{\pi} \sim 20$ V and a conversion bandwidth of ~ 2.6 nm. The splitting ratios of the adiabatic coupler PBS in the IPC device are $>99\%$ for both polarization modes over a broad spectral range from 1500–1640 nm. The EO mode of the implemented IPC device is activated when the PPLN EO PMC section is driven by an external voltage; the characterized EO switching/power division behavior of the device is in good agreement with the theoretical fit. The tunability of the EO IPC device in the 100-nm experimental spectral range is also demonstrated via the temperature tuning. The featured broad tunability and high integrability of the EO device presented in this study facilitates it to be an advantageous building block for realizing an on-chip photonic system.

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1. Introduction

Integrated photonic circuits (IPC) have emerged as a key technology for demonstrating state-of-the-art results in many interesting areas such as the optical communication networks and the quantum photonics [1–3]. Various passive and active optical elements have been developed as building blocks for realizing IPC of versatile applications. For example, IPC technology has been widely used in the implementation of photonic quantum circuits (PQC) where (heterogeneous) building blocks like qubit sources, linear optical elements, modulators, and single-photon detectors are aimed to be integrated on a common substrate [4]. In such a system, linear optical elements have been used in building a photonic circuit for the creation, manipulation, and test of (multidimensional) quantum entanglement states (qudits) from the generated qubits [5] as well as in implementing a reprogrammable logic gate circuitry for quantum computing [6]. Among the elements, waveguide couplers have been the fundamental ones in the construction of a variety of functional devices such as beam/mode splitters, switches, (de)multiplexers, interferometers, and modulators [7,8]. Being a basic building block, its working/operating specifications certainly determine the performance and practicability of the developed IPC. Conventional directional couplers usually suffer from the low fabrication tolerance and limited operating bandwidth due to the mode interference nature they work on.

The use of a fabrication tolerant and broad working-bandwidth (with respect to such as wavelength and temperature) fundamental optical element (such as a waveguide coupler) is

thus crucial to increase the scalability and fidelity of the built IPC. Moreover, in terms of the integrability, a tolerant and broadband element also facilitates its on-chip integration with other specific functional elements even though they feature a narrower operating bandwidth. Besides mode interference devices, many important (active) photonic components functioning based on the access of the material nonlinearity also work only for a limited operating (such as temperature, wavelength, or voltage) bandwidth subject to their based phase-matching conditions. In PQC, quadratic nonlinearity has been used to implement efficient qubit sources [9] and fast switches, modulators, mode converters, etc. [10]. A high-speed (>GHz) manipulation of the generated qubits (on phase, polarization, etc.) can be further expected when the qubit source operates with an electro-optic (EO) device.

Lithium niobate (LN) is an iconic photonic material featuring high quadratic nonlinearity and efficient EO effects [11]. Specifically, while assessing the off-diagonal EO tensor element r_{51} of LN, an active polarization rotator can be implemented in a periodically poled domain structure with the rotation angle (θ_{ro}) being a function of the strength of the applied electric field E_y along the crystallographic y axis [12]. Accordingly, a polarization mode converter (PMC) can be realized when such a polarization rotator operates at $\theta_{ro}(E_y) = 90^\circ$. We have successfully demonstrated a fast modulation of the laser operation and a fast manipulation/switching of the spectrum of an optical parametric oscillator by building the unique EO polarization mode conversion mechanism in the intracavity integrated domain-engineered LN wavelength converters [13,14]. In view of these demonstrated excellent integrability even with bulk components, we have further conceptually proposed in our previous study a fully integrated quantum polarization state source where a periodically poled LN (PPLN) EO PMC is integrated with a type-II phase-matched PPLN spontaneous parametric down converter (SPDC) and an adiabatic-coupler based *wavelength-dependent* polarization beam splitter (PBS) for on-chip quantum-polarization state preparation [15]. Later on, a similar on-chip scheme where a PPLN EO PMC and a segmented PPLN EO PMC have been used to perform a uniquely dynamic time management in a type-II PPLN SPDC source via the EO manipulation of the generated photon (polarization) states when they operate at the input and one of the output ports of a directional-coupler based PBS, respectively, has been realized [16]. However, such a scheme has to operate at a specific temperature for obtaining optimum performance due to the monolithic integration of two narrowband quasi-phase-matching (QPM) devices (i.e., PPLN SPDC and PPLN EO PMC). It is even critical that the two devices actually further work with a PBS implemented using a directional coupler which is usually sensitive to the wavelength and fabrication errors [17].

A fabrication tolerant, broadband integrated-optical PBS has been demonstrated in LN waveguide couplers [18] in a counterintuitive light transfer scheme, achieved by classically emulating the stimulated Raman adiabatic passage (STIRAP) process observed in a quantum system [19]. In this study, we further demonstrate the integration of a PPLN EO PMC with a broadband adiabatic-passagge optical-coupler based PBS on a LN photonic (waveguide) circuit chip to function as a fast optical switch or, more in general, an EO controllable power divider. Such a novel device features broadband tunability and therefore high integrability. Besides being an EO switch, this device can be an advantageous integrated building block of PQC for implementing various interesting quantum devices such as a postselection-free polarization-entangled photon-pair transmitter [20], a cross time-bin-entangled photon emitter [21], and a timing management and quantum state manipulation device [16,22].

2. Device design and simulation

In this demonstration, we study a unique device integrating two photonic elements of important optical functionalities which are the polarization rotation/polarization mode conversion and the polarizing beam splitting. Figure 1 shows the schematic of such an IPC device with the first section being a 1-cm long PPLN waveguide of a domain grating period $\Lambda = 23 \mu\text{m}$ and

the second section being a 3.3-cm long five-waveguide adiabatic coupler. The whole device architecture is designed based on the dispersion of a Ti-diffused LN (Ti:LN) waveguide [23]. The PPLN performs the power conversion between the TE and TM polarization modes in the waveguide, enabled by the periodic sign modulation of the EO coefficient r_{51} in space along the crystallographic x axis in the presence of E_y . The conversion efficiency of such an EO PMC can be expressed as $\eta = \sin^2(|\kappa|L)$, where L is the device length and $|\kappa| = \zeta(\pi/\lambda_0)n_{o,eff}^{3/2}n_{e,eff}^{3/2}|r_{51}E_y|\vartheta$ with ζ being the Fourier reduction factor ($\equiv 2\sin(\pi D)/\pi$), λ_0 being the working wavelength, $n_{o,eff}$ and $n_{e,eff}$ being the effective refractive indices of the ordinarily (TE) and extraordinarily (TM) polarized modes, respectively, and ϑ being the overlap efficiency of the two polarization-mode fields and the external field (E_y), for a 1st-order PPLN grating with a domain duty cycle D (ratio of the length of the reversed domain to the grating period Λ of the PPLN) [12]. Accordingly, the polarization direction of a (linearly polarized) wave can be varied in the device via tuning E_y . The voltage, V_π , required for switching the polarization states between the TE and TM modes can be derived as

$$V_\pi = \pi \frac{\lambda_0}{4} \frac{1}{r_{51} \sin(\pi D) n_{o,eff}^{3/2} n_{e,eff}^{3/2}} \frac{d}{L} \frac{1}{\vartheta}, \quad (1)$$

where d is the electrode spacing for the E_y application. From Eq. (1), a switching voltage of such a Ti:PPLN EO PMC can be as low as $V_\pi \sim 5.5$ V when $\lambda_0 = 1.55$ μm , $d = 12$ μm , and $L = 10$ mm are used under the ideal domain structure and maximum overlap efficiency (i.e., $D = 0.5$ and $\vartheta = 1$).

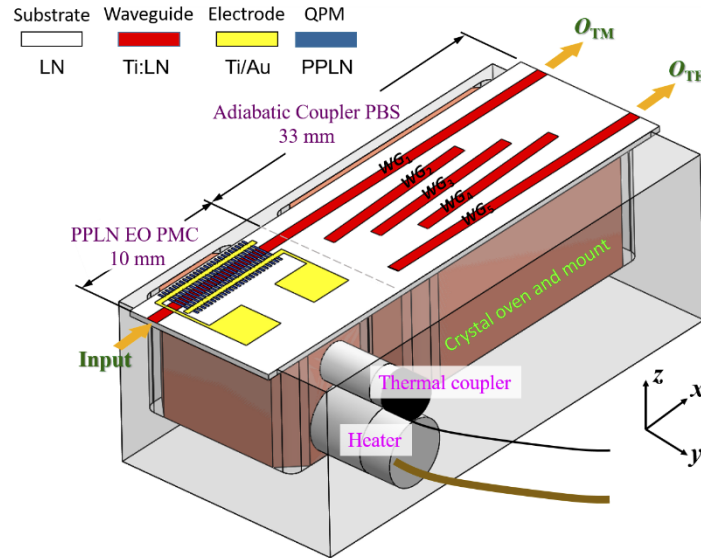


Fig. 1. Schematic of the IPC device integrating a 1-cm long PPLN EO PMC and a 3.3-cm long five-waveguide adiabatic coupler on a Ti:LN photonic-circuit chip to function as an EO switch/EO controllable power divider.

The EO modulated wave will directly enter the five-waveguide adiabatic coupler section. In the design, the resultant TM- and TE-mode components of the wave after interacting with the PPLN section will go straight through WG_1 (“Bar” state) and cross over to WG_5 (“Cross” state) via the adiabatic tunneling effect, respectively, and exit from the output ports O_{TM} and O_{TE} of the adiabatic coupler system. The five-waveguide scheme thus functions as a photonic-circuit PBS, in which the whole integrated device (Fig. 1) can work as a power divider with a power ratio being tunable via the EO control of the wave polarization state through the upstream PPLN PMC section

and become an EO switch when the operating voltages of the PPLN PMC are switched between 0 V and V_π for either a TE or a TM polarized input wave. In the design of the five-waveguide PBS, the three intermediate waveguides, corresponding to the intermediate dressed states in a STIRAP quantum system [24], have been employed to engineer a proper adiabaticity (corresponding to the resonance condition between the intermediate dressed states and the initial and final states of the quantum system) to enable a high-efficiency adiabatic passage of light from WG_1 to WG_5 without a significant excitation of the intermediate waveguides for the TE polarization mode, while the adiabaticity will forbid the power transfer of the TM polarization mode (from WG_1) to any other waveguides. Such a highly polarization-dependent adiabaticity can't be achieved in a simpler 3-waveguide Ti:LN adiabatic coupler with only a single intermediate waveguide due to the relative small birefringence ($\Delta n = n_{o,eff} - n_{e,eff} \sim 0.07$ around the 1.55- μm band) of the material. We have demonstrated a high-extinction-ratio PBS in a stand-alone five-waveguide architecture in LN with an effective coupling length of 3 cm (which is determined by the length of the three intermediate waveguides) [18]. In this study, as shown in Fig. 1, the adiabatic coupler PBS is designed to be further integrated with a 1-cm long PPLN EO device on a chip. Therefore, a more careful and optimized design has been developed here to further reduce the length of the five-waveguide adiabatic coupler device, which is desirable to keep the integrated device reasonably compact. Inspired by a theoretical study of multistate STIRAP-like atomic systems reported in [25] where it shows an adiabatic population transfer can be realized in such a system via the application of a properly arranged counterintuitive sequence of pump, Stokes, and intermediate pulses (i.e., the transfer behavior depends on the timing relationship among the pulses), we resort to the design of a configuration of the three intermediate passage waveguides for reaching a shorter coupler length while obtaining a reasonably high adiabaticity. The process involves the derivation of structure parameters (relative location and spacing) of the waveguides, which is equivalent to the engineering of a relationship of spatially dependent coupling strength (which is a function of the coupling coefficients) among the waveguides to emulate the use of a proper timing sequence and strong enough strength (Rabi frequency) of the exciting pulses in the aforementioned STIRAP-like atomic systems.

Figure 2 shows the calculated coupling coefficients of the TE mode between the waveguides WG_i and WG_j of the proposed five-waveguide adiabatic coupler scheme, $\kappa_{ij}(x) (\equiv \pi/2l_{c0}e^{S_{ij}(x)/r})$, where l_{c0} and r are waveguide characteristic parameters whose values can be experimentally determined [17]), as a function of the propagation distance along the x direction. Table 1 lists the waveguide structure parameters (refer to Fig. 2) used in this calculation. Uniquely in this new design, the relative positions of the three equal-length ($=l_p$), equally-spaced (with a spacing $S_{23}(x)=S_{34}(x)$). Here we define $S_{ij}(x_k)$ as the edge-to-edge distance between WG_i and WG_j at x_k) slash passage waveguides have been shifted by an amount Δd along the x coordinate, leading to a high-adiabaticity counterintuitive light transfer between x_3 and x_4 in which $\kappa_{45}(x_3) > \kappa_{12}(x_3)$ and $\kappa_{23}(x) = \kappa_{34}(x) \geq (\kappa_{45}(x_3) \text{ and } \kappa_{12}(x_4))$ can be attained with the former (spatially dependent coupling coefficients) relationship satisfying the adiabatic condition [26] and the latter one being the spatial realization of three important criteria concluded in the timing excitation of a STIRAP-like atomic system [25]: (1) Strong enough intermediate pulses (spatial correspondence: a large $\kappa_{23}(x)$ and $\kappa_{34}(x)$) not only minimize the intermediate populations (spatial correspondence: minimal power excited in the intermediate waveguides), but also increase adiabaticity. (2) Intermediate pulses are found to be most appropriate when their strengths are of the same order (spatial correspondence: $\kappa_{23}(x) \sim \kappa_{34}(x)$) and (3) they arrive simultaneously with or earlier than the Stoke pulse and terminate simultaneously with or later than the pump pulse (spatial correspondence: $\kappa_{23}(x) \sim \kappa_{34}(x) \geq \kappa_{45}(x_3) \text{ and } \kappa_{12}(x_4)$).

The structure parameters listed in Tab. 1, derived using the "Beam Propagation Method" (BPM) [27], are one representative set for constructing a five-waveguide adiabatic coupler to meet the high-adiabaticity light transfer conditions discussed above for the TE mode while

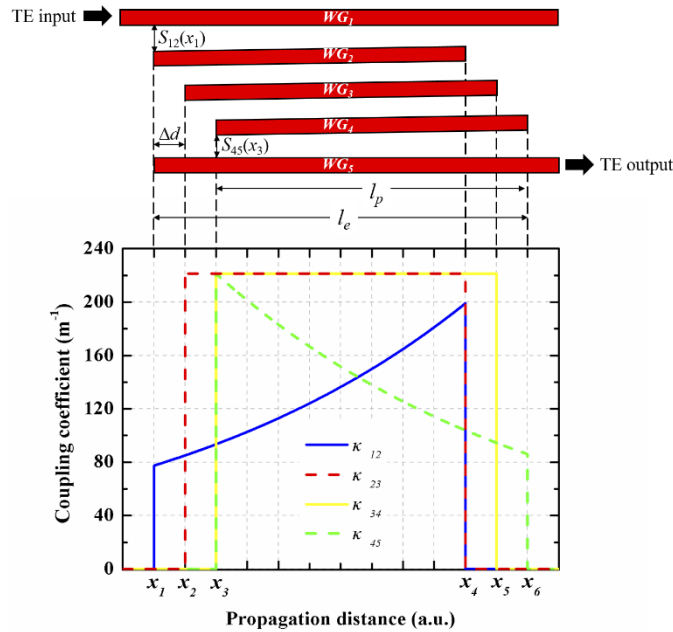


Fig. 2. Calculated coupling coefficients $\kappa_{ij}(x)$ between the waveguides WG_i and WG_j of the five-waveguide system with structure parameters as listed in Tab. 1 as a function of the propagation distance along the x direction.

Table 1. Structure parameters (see text for their definitions) adopted in the simulation and fabrication of the five-waveguide adiabatic coupler working as a PBS in the proposed IPC device

Key structure parameters (μm)							
Waveguide dimension			Waveguide spacing				Shift of passage waveguides
w_c	l_p	l_e	$S_{12}(x_1)$	$S_{23}(x)$	$S_{24}(x)$	$S_{45}(x_3)$	Δd
7.3	20000	24000	12.7	8	8	8	2000

simultaneously preventing the TM mode from coupling to any other waveguides. Such a highly polarization-mode splitting condition can be found in a range of Δd from 1.8 to 2.25 mm while keeping other structure parameters fixed, showing a high structural tolerance as expected from the feature of an adiabatic coupler. Figure 3(a) shows the simulated evolutions of the wave intensity in the five-waveguide adiabatic coupler system (the leftmost is again a system scheme) based on the structure parameters listed in Tab. 1 for the TE- and TM-polarized 1550-nm fundamental modes initially excited in WG_1 , respectively. The results clearly reveal the signature of the optical adiabatic tunneling process in the proposed waveguide system where a great majority of the input power has transferred from WG_1 to WG_5 in a counterintuitive coupling manner without significant excitation of intermediate waveguides WG_2 , WG_3 , and WG_4 for the TE mode. A coupling efficiency of $>99.3\%$ can be obtained with such a coupler according to the simulation. The coupling efficiency is defined as $\eta = P_5/P_t$, where P_5 and P_t are the power calculated/measured at WG_5 output port (O_{TE}) and all the output ports ($P_t = P_1 + P_5$ for our coupler scheme), respectively. On the other hand, only $\sim 0.03\%$ of the input TM-mode power will transfer to WG_5 . Accordingly, the polarization extinction ratios (PER) calculated with this PBS scheme are ~ 35.2 and ~ 39.0 dB for the TE mode (at O_{TE}) and for the TM mode (at O_{TM}) at 1550 nm, respectively. The PER of a PBS is defined as $r_{TE} = 10 \log(P_5^{TE}/P_5^{TM})$ and $r_{TM} = 10 \log(P_1^{TM}/P_1^{TE})$ for TE- and TM-polarized

modes, respectively, which can be obtained with an input wave in the 45° polarization state. Remarkably, when compared to that studied in [18], the effective coupling length ($l_e = 24$ mm) of this newly designed (based on the spatially engineered coupling strength method proposed above) adiabatic waveguide PBS has been successfully reduced by 20%, while having comparable device performance.

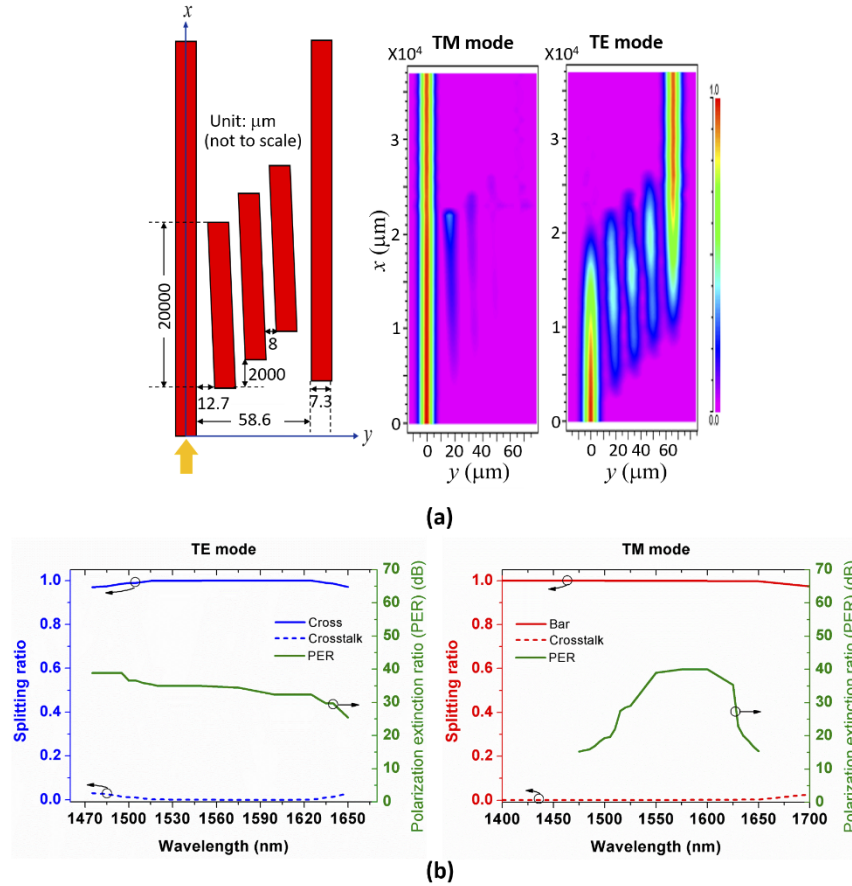


Fig. 3. (a) Simulated evolutions of the wave intensity in the five-waveguide adiabatic coupler system based on the structure parameters listed in Tab. 1 for the TE- and TM-polarized 1550-nm fundamental modes initially excited in WG_1 . (b) Calculated splitting ratios and PER of the five-waveguide PBS as a function of the working wavelength for both TE- and TM-polarized fundamental modes. The legend “Crosstalk” denotes the wave monitored at O_{TM} and O_{TE} in the “Cross” and “Bar” states, respectively.

Besides, the adiabatic coupler PBS studied in this work exhibits it is also broadband operational. Figure 3(b) shows the calculated splitting ratios of the five-waveguide PBS (Fig. 2/ Fig. 3(a)) as a function of the working wavelength for both polarization modes. The results present the device can be with a broad bandwidth of ~ 140 nm (1500–1640 nm) and >250 nm (1400–1650 nm) for the TE- and TM-polarized fundamental modes, respectively, at a high splitting ratio of $>99\%$. The corresponding PER (green lines in Fig. 3(b)) of the five-waveguide PBS are found to be ≥ 20 dB over the 1500–1640-nm region for both polarization modes as well, which again marks the broadband, high polarizing characteristics of our proposed device. This important feature of the PBS facilitates its integration in the unique photonic device proposed in Fig. 1 with the PPLN EO PMC which is characterized to have a limited bandwidth (~ 2.6 nm·cm at 1.5- μm band, see

below) subject to the based QPM condition, and such an integrability is significant to realize a broadband tunable EO building block (specifically an EO switch/power divider here) in (LN) photonic circuit platforms.

3. Device fabrication and performance characterization

We fabricated the IPC device, composing of a PPLN EO PMC and an adiabatic coupler PBS as schematically shown in Fig. 1, on a 43-mm-long (along the crystallographic x axis) z -cut LN chip of a thickness 0.5 mm. The 1-cm long PPLN section has a domain grating period of 23 μm . The 23- μm domain period is used to quasi-phase-match the polarization mode conversion process of a 1550 nm wave in Ti:LN at 134°C. A higher operating temperature (rather than at room temperature) is employed to reduce the possible damage from the photorefractive effect, though we did not observe this effect in the experiment even at near room temperature. The five-waveguide system in the second device section has been constructed based on the structure parameters listed in Tab. 1. The waveguide architecture of the IPC was first defined by a Ti strip layer of a thickness of 100 nm on the $-z$ surface of the LN chip. The Ti:LN waveguides were then formed by conducting the thermal indiffusion process [28] on the sample in a 3-zone furnace at 1035°C for 12 hours. The refractive index profiles of the fabricated waveguides are characterized to have surface index changes of $\Delta n_o = 0.0086$ and $\Delta n_e = 0.0157$ and e^{-1} depths of $d_o = 5.8 \mu\text{m}$ and $d_e = 4.5 \mu\text{m}$ (the subscripts o and e denote quantities associated to the ordinarily (TE) and extraordinarily (TM) polarized modes, respectively) using a prism coupler at 532 nm, supporting the guiding of single TE and TM fundamental modes in a channel waveguide of width $w_c = 7.3 \mu\text{m}$ in the spectral range of interest, 1400–1700 nm. The periodic domain structure was then produced in the first device section by using the standard electric-field poling technique [29] after the surface polish of the $+z$ face of the LN chip to remove the domain-inverted layer formed during the thermal indiffusion process [30]. The end faces of the chip were then optically polished without applying any optical coating to them. The device fabrication was accomplished after the coating of the strip electrodes (Ni/Au with thicknesses of 50 and 150 nm for the Ni and Au layers, respectively) on both sides of the PPLN EO PMC waveguide with a gap of 12 μm between them for the voltage (V_{EO}) application to build the field E_y as discussed in section 2 (see also Fig. 1). Figure 4 shows the microscopic images of the $-z/+z$ surfaces of the fabricated IPC device in portions of the PPLN EO PMC (PPLN domain structure was revealed by HF etching on the $+z$ surface of the device in order not to damage the waveguides fabricated on the $-z$ surface) and the adiabatic coupler PBS.

We then demonstrate the implemented LN IPC device to work as, but not limited to, a broadband tunable EO switch/EO controllable power divider.

The performance of the LN IPC device was characterized in an optical testbed using a linearly polarized, tunable (1495–1620 nm) external cavity laser as the light source. The laser is fiber coupled and polarization controlled (using a 3-paddle polarization controller (3PPC)) to excite a TE or TM mode in the input waveguide of the IPC chip through the butt coupling scheme. The input power level is ~ 1 mW. The chip was installed in a temperature controlled (room temperature to 150°C) crystal oven which was mounted on a multi-axis precision translation stage. A pair of fine needle probes were used to contact the two side electrodes of the PPLN EO PMC section, respectively, through which external voltages (V_{EO}) were supplied. An optical detection system consisting of an objective lens, an iris, a polarizer, a photodetector, and an infrared CCD was used to measure and analyze the polarization state, power, and intensity profile of an output mode of the IPC device. The propagation losses of the waveguides were around 0.4 and 0.2 dB/cm for the TE and TM modes at the 1550 nm band, characterized using the Fabry-Perot method [31]. Besides, the insertion losses of the fabricated IPC chip were measured to be 5.44 and 6.19 dB for the TE and TM modes at 1550 nm, respectively, from which we estimate the total coupling losses (for both input and output couplings) for our device to be ~ 3.72

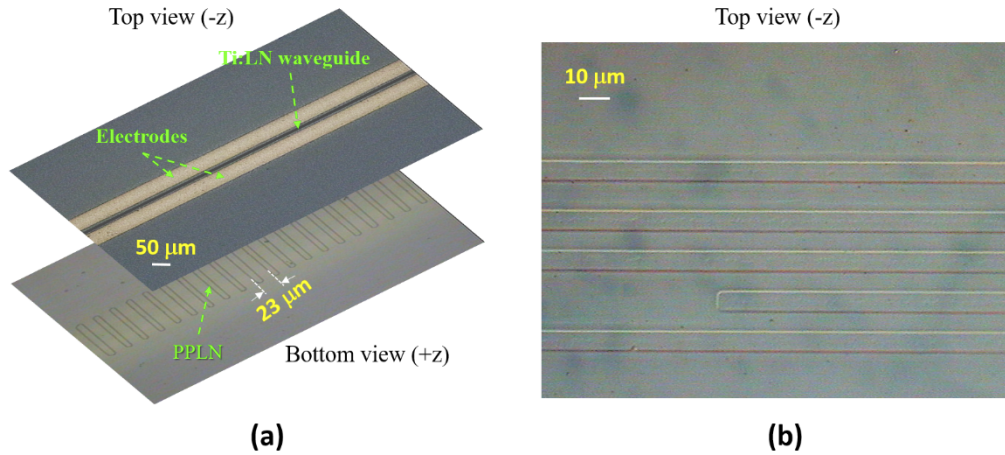


Fig. 4. Microscopic images of the $-z/+z$ surfaces of the fabricated IPC device in portions of (a) the PPLN EO PMC and (b) the adiabatic coupler PBS.

and ~ 5.33 dB for the TE and TM modes, respectively. The coupling loss can be further reduced by optimizing the mode-matching scheme, improving the end-facet polishing quality, and doing the fiber pig-tailing. We then first characterized the device operating in the passive mode (i.e., when $V_{EO} = 0$). Figure 5 shows the measured splitting ratios of the IPC device as a function of the laser wavelength with the polarization being controlled to excite TE- (blue dots and circles for “cross” and “crosstalk” couplings, respectively) and TM- (red solid and open diamonds for “bar” and “crosstalk” couplings, respectively) modes, respectively, in the input waveguide (WG_1) using the 3PPC under $V_{EO} = 0$. It shows the measured polarization splitting spectra of the device are in good agreement with the calculated results (see Fig. 3(b)) over a broad band spanning the telecom S, C, and L bands, inferring a high fabrication tolerance of the five-waveguide PBS as expected for an adiabatic coupler.

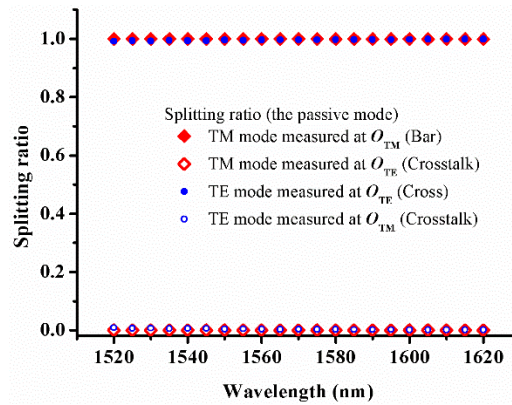


Fig. 5. Measured splitting ratios of the IPC device as a function of the laser wavelength for the TE (blue dots and circles) and TM (red solid and open diamonds) polarization modes excited in WG_1 under $V_{EO} = 0$ (the passive mode).

The EO mode or the full operation mode of the developed IPC device is activated when the PPLN EO PMC is driven by an external voltage. Figure 6(a) shows the characterized EO switching behavior of the IPC device for an input wave with the TM-polarized fundamental

mode at 1550 nm at 134°C. The data points in the figure (green dots) were measured in terms of the coupling efficiency (as defined above) of the adiabatic coupler PBS in the IPC device. The switching voltage, V_{π} , is found to be ~ 20 V from the measurement, implying an overlap efficiency (ϑ) of ~ 0.34 according to Eq. (1) with a PPLN domain duty cycle $D \sim 30\%$ revealed from the image shown in Fig. 4(a). With the deduced ϑ , we theoretically fit the voltage tuning curve of the IPC device, as shown in Fig. 6(a) (orange line). Figure 6(b) shows the captured images of the output spatial modes from the IPC device when operating at $V_{EO} = 0, 10$, and 20 V, respectively. These results clearly exhibit that the implemented device can function as an EO switch and an EO controllable power divider via switching and tuning the V_{EO} between 0 and 20 V, respectively. The power switching efficiency (the ratio of the power measured at O_{TE} at 20 V to that measured at O_{TM} at 0 V for a TM-polarized input wave) is $>99\%$ according to the result obtained in Fig. 6(b). To test the fast switching characteristics of this unique EO device, we drove the PPLN PMC section of the IPC with 1-kHz, 40-V (-20 V to $+20$ V) peak-to-peak (V_{p-p}) square and triangle voltage waveforms (generated from a function generator available in our lab). Figures 7(a) and 7(b) show the time responses of the EO IPC device at μ s and sub-ms switching speeds, respectively. The results reveal our device can well respond fast ($\sim$ MHz) switching voltage signals with a good linear responsivity. A higher switching speed ($>$ GHz) of the device can be expected when a careful electrode design is adopted [32]. Besides its remarkable value in integrated photonics, such a highly integrated fast polarization rotator and broadband PBS device can also be of great interest to the photonic quantum applications using especially polarization entangled/encoded qubits as discussed in Introduction section.

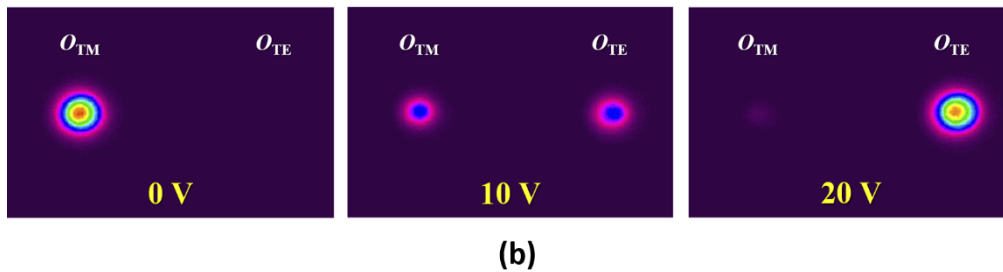
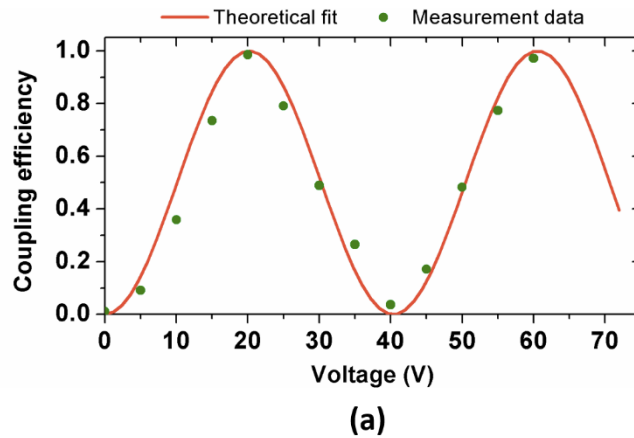


Fig. 6. (a) Measured (green dots) and calculated (orange line) coupling efficiency of the IPC device as a function of the driving voltage V_{EO} for a TM polarization beam at 1550 nm at 134°C. (b) Output spatial mode images of the IPC device captured by an infrared CCD when operating at $V_{EO} = 0, 10$, and 20 V, respectively.

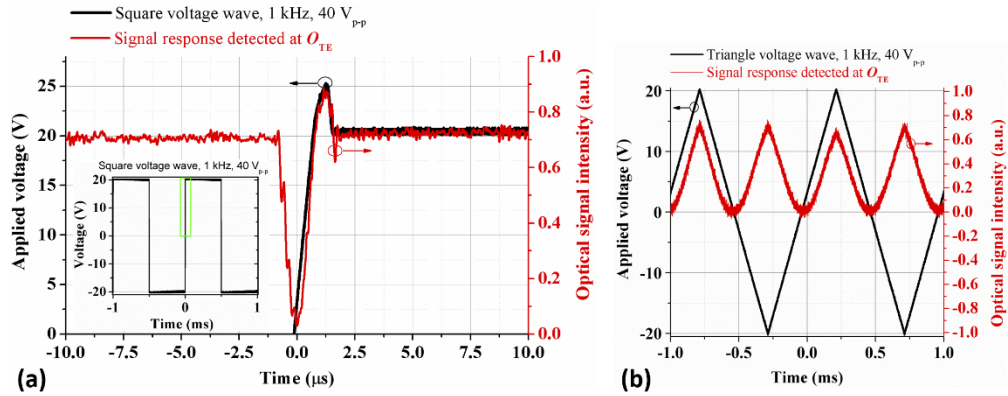


Fig. 7. Time responses of the EO IPC device when driven by 1-kHz, 40-V (-20 V to $+20$ V) peak-to-peak (V_{p-p}) (a) square and (b) triangle voltage waveforms. The inset in (a) shows the complete square waveform of the driving voltage source. The portion of the square waveform (black curve) displayed in (a) (μ s scale) corresponds to the scope defined by the green box in the inset.

Figure 8 shows the measured (solid dots) and calculated (solid lines) EO switching spectra of the IPC device when tuned by temperature under $V_{EO} = 20$ V for TM-polarized input waves. The spectra are basically identical to the wavelength tuning curves of the PPLN EO PMC described by the based QPM condition. The measured tuning curves are in good agreement with the theoretical fits and reveal a bandwidth of ~ 2.6 nm. Fortunately, this limited operating bandwidth of the PMC does not limit the tunability of the IPC device thanks to the broadband characteristics of the adiabatic coupler PBS (see Fig. 5). The experimental data acquisition range was limited to the working ranges of the laser and crystal oven; the actual tuning range of the device is expected to be as broad as the simulated results shown in Fig. 3(b). The obtained tuning rate of the IPC device is 0.84 nm/ $^{\circ}$ C. Such a tunability facilitates the IPC device realized in this study to be a more versatile and compatible building block for implementing an on-chip photonic system, which can't be achieved in an IPC when the PMC is integrated with a conventional directional-coupler based PBS.

The IPC architecture studied in this work can be further implemented in a thin-film LN on insulator platform supporting nanometric circuit structuring based on the state-of-the-art fabrication technology [33,34], in which the footprint and the working voltage of the device can be largely reduced [35].

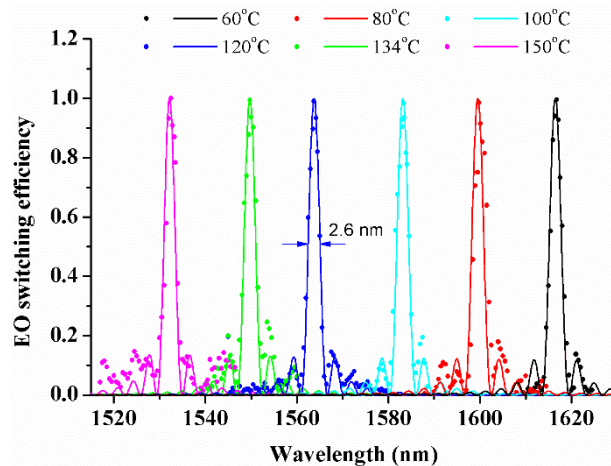


Fig. 8. Measured (solid dots) and calculated (solid lines) EO switching spectra of the IPC device when tuned by temperature under $V_{EO} = 20$ V for TM-polarized input waves. The experimental data acquisition range was limited to the working ranges of the laser and crystal oven.

4. Conclusion

We have proposed and realized a unique IPC device functioning as a broadband tunable EO switch as well as an EO controllable power divider in a Ti:LN waveguide platform. The device integrates a 1-cm long, 23- μm domain-period PPLN working as an EO PMC and an effectively 2.4-cm long, five-waveguide adiabatic-coupler based PBS on a 43-mm long LN chip. The PPLN section performs the EO active conversion between the TE and TM modes with a half-wave voltage of $V_{\pi} \sim 20$ V, while the adiabatic coupler section permits a broadband polarization mode splitting with a high splitting ratio ($>99\%$), achieved based on a careful engineering of the spatially dependent coupling strength of the waveguide system. The EO switching/power division ability of the developed IPC device has been demonstrated via the switching/tuning of the voltage (V_{EO}) applied to the PPLN section between 0 and 20 V. The tunability of the novel integrated device has also been achieved across the telecom S, C, and L bands using the temperature with a tuning rate of 0.84 nm/ $^{\circ}\text{C}$. The IPC studied in this work can also be of great value to expedite the development of integrated photonic quantum technology.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

References

1. X. Chen, M. M. Milosevic, S. Stanković, S. Reynolds, T. D. Bucio, K. Li, D. J. Thomson, F. Gardes, and G. T. Reed, "The emergence of silicon photonics as a flexible technology platform," *Proc. IEEE* **106**(12), 2101–2116 (2018).
2. W. Bogaerts, D. Perez, J. Capmany, D. A. B. Miller, J. Poon, D. Englund, F. Morichetti, and A. Melloni, "Programmable photonic circuits," *Nature* **586**(7828), 207–216 (2020).
3. D. Pérez, I. Gasulla, P. D. Mahapatra, and J. Capmany, "Principles, fundamentals, and applications of programmable integrated photonics," *Adv. Opt. Photon.* **12**(3), 709 (2020).
4. J. Wang, F. Sciarrino, A. Laing, and M. G. Thompson, "Integrated photonic quantum technologies," *Nat. Photonics* **14**(5), 273–284 (2020).

5. J. Wang, S. Paesani, Y. Ding, R. Santagati, P. Skrzypczyk, A. Salavrakos, J. Tura, R. Augusiak, L. Mančinska, D. Bacco, D. Bonneau, J. W. Silverstone, Q. Gong, A. Acín, K. Rottwitt, L. K. Oxenløwe, J. L. O'Brien, A. Laing, and M. G. Thompson, "Multidimensional quantum entanglement with large-scale integrated optics," *Science* **360**(6386), 285–291 (2018).
6. X. Qiang, X. Zhou, J. Wang, C. Wilkes, T. Loke, S. O'Gara, L. Kling, G. Marshall, R. Santagati, T. Ralph, J. Wang, J. O'Brien, M. Thompson, and J. Matthews, "Large-scale silicon quantum photonics implementing arbitrary two-qubit processing," *Nat. Photonics* **12**(9), 534–539 (2018).
7. R. V. Schmidt and R. C. Alfiness, "Directional coupler switches, modulators, and filters using alternating $\Delta\beta$ techniques," *IEEE Trans. Circuits Syst.* **26**(12), 1099–1108 (1979).
8. H. C. Cheng and R. V. Ramaswamy, "Symmetrical directional coupler as a wavelength multiplexer-demultiplexer: theory and experiment," *IEEE J. Quantum Electron.* **27**(3), 567–574 (1991).
9. A. Martin, A. Issautier, H. Herrmann, W. Sohler, D. B. Ostrowsky, O. Alibart, and S. Tanzilli, "A polarization entangled photon-pair source based on a type-II PPLN waveguide emitting at a telecom wavelength," *New J. Phys.* **12**(10), 103005 (2010).
10. P. R. Sharapova, K. H. Luo, H. Herrmann, M. Reichelt, T. Meier, and C. Silberhorn, "Toolbox for the design of LiNbO₃-based passive and active integrated quantum circuits," *New J. Phys.* **19**(12), 123009 (2017).
11. L. Arizmendi, "Photonic applications of lithium niobate crystals," *phys. stat. sol. (a)* **201**(2), 253–283 (2004).
12. C. Y. Huang, C. H. Lin, Y. H. Chen, and Y. C. Huang, "Electro-optic Ti:PPLN waveguide as efficient optical wavelength filter and polarization mode converter," *Opt. Express* **15**(5), 2548 (2007).
13. Y. H. Chen, W. K. Chang, C. L. Chang, and C. H. Lin, "Single Aperiodically Poled Lithium Niobate for Simultaneous Laser Q-switching and Second-Harmonic Generation in a 1342-nm Nd:YVO₄ Laser," *Opt. Lett.* **34**(11), 1711–1713 (2009).
14. T. J. Wang, L. M. Deng, H. P. Chung, W. K. Chang, T. D. Pham, Q. H. Tseng, R. Geiss, T. Pertsch, and Y. H. Chen, "Electro-optically spectrum switchable, multi-wavelength optical parametric oscillators based on aperiodically poled lithium niobate," *Opt. Lett.* **45**(20), 5848–5851 (2020).
15. H. P. Chung, K. H. Huang, K. Wang, S. L. Yang, S. Y. Yang, C. I. Sung, A. S. Solntsev, A. A. Sukhorukov, D. N. Neshev, and Y. H. Chen, "Asymmetric adiabatic couplers for fully-integrated broadband quantum-polarization state preparation," *Sci. Rep.* **7**(1), 16841 (2017).
16. K. H. Luo, S. Brauner, C. Eigner, P. R. Sharapova, R. Ricken, T. Meier, H. Herrmann, and C. Silberhorn, "Nonlinear integrated quantum electro-optic circuits," *Sci. Adv.* **5**(1), eaat1451 (2019).
17. R. C. Alfiness, R. V. Schmidt, and E. H. Turner, "Characteristics of Ti-diffused lithium niobate optical directional couplers," *Appl. Opt.* **18**(23), 4012–4016 (1979).
18. H. P. Chung, C. H. Lee, K. H. Huang, S. L. Yang, K. Wang, A. S. Solntsev, A. A. Sukhorukov, F. Setzpfandt, and Y. H. Chen, "Broadband on-chip polarization mode splitters in lithium niobate integrated adiabatic couplers," *Opt. Express* **27**(2), 1632–1645 (2019).
19. K. Bergmann, H. Theuer, and B. W. Shore, "Coherent population transfer among quantum states of atoms and molecules," *Rev. Mod. Phys.* **70**(3), 1003–1025 (1998).
20. T. Suhara, "Generation of quantum-entangled twin photons by waveguide nonlinear-optic devices," *Laser & Photon. Rev.* **3**(4), 370–393 (2009).
21. A. Martin, F. Kaiser, A. Vernier, A. Beveratos, V. Scarani, and S. Tanzilli, "Cross time-bin photonic entanglement for quantum key distribution," *Phys. Rev. A* **87**(2), 020301 (2013).
22. F. Kaiser, A. Issautier, L. A. Ngah, O. Dănilă, H. Herrmann, W. Sohler, A. Martin, and S. Tanzilli, "High-quality polarization entanglement state preparation and manipulation in standard telecommunication channels," *New J. Phys.* **14**(8), 085015 (2012).
23. S. Fouchet, A. Carencu, C. Daguet, R. Guglielmi, and L. Riviere, "Wavelength dispersion of Ti induced refractive index change in LiNbO₃ as a function of diffusion parameters," *J. Lightwave Technol.* **5**(5), 700–708 (1987).
24. G. D. Valle, M. Ornigotti, T. T. Fernandez, P. Laporta, S. Longhi, A. Coppa, and V. Foglietti, "Adiabatic light transfer via dressed states in optical waveguide arrays," *Appl. Phys. Lett.* **92**(1), 011106 (2008).
25. N. V. Vitanov, "Adiabatic population transfer by delayed laser pulses in multistate systems," *Phys. Rev. A* **58**(3), 2295–2309 (1998).
26. H. P. Chung, K. H. Huang, S. L. Yang, W. K. Chang, C. W. Wu, F. Setzpfandt, T. Pertsch, D. N. Neshev, and Y. H. Chen, "Adiabatic light transfer in titanium diffused lithium niobate waveguides," *Opt. Express* **23**(24), 30641–30650 (2015).
27. J. Van Roey, J. van der Donk, and P. E. Lagasse, "Beam-propagation method: analysis and assessment," *J. Opt. Soc. Am.* **71**(7), 803–810 (1981).
28. W. K. Burns, P. H. Klein, E. J. West, and L. E. Plew, "Ti diffusion in Ti:LiNbO₃ planar and channel optical waveguides," *J. Appl. Phys.* **50**(10), 6175–6182 (1979).
29. L. E. Myers, R. C. Eckardt, M. M. Fejer, R. L. Byer, W. R. Bosenberg, and J. W. Pierce, "Quasi-phase-matched optical parametric oscillators in bulk periodically poled LiNbO₃," *J. Opt. Soc. Am. B* **12**(11), 2102–2116 (1995).
30. G. Schreiber, H. Suche, Y. L. Lee, W. Grundkoetter, V. Quiring, R. Ricken, and W. Sohler, "Efficient cascaded difference frequency conversion in periodically poled Ti:LiNbO₃ waveguides using pulsed and cw pumping," *Appl. Phys. B* **73**(5-6), 501–504 (2001).

31. R. Regener and W. Sohler, "Loss in low-finesse Ti:LiNbO_3 optical waveguide resonators," *Appl. Phys. B* **36**(3), 143–147 (1985).
32. D. Yang, Y. Chen, M. Xiang, Y. Wang, and P. Liu, "Traveling wave electrode design for a LiNbO_3 integrated optical switch," *Proc. SPIE* **11334**, 113341B (2019).
33. M. Younesi, R. Geiss, S. Rajaei, F. Setzpfandt, Y. H. Chen, and T. Pertsch, "Periodic poling with a micrometer-range period in thin-film lithium niobate on insulator," *J. Opt. Soc. Am. B* **38**(3), 685–691 (2021).
34. D. Zhu, L. Shao, M. Yu, R. Cheng, B. Desiatov, C. J. Xin, Y. Hu, J. Holzgrafe, S. Ghosh, A. Shams-Ansari, E. Puma, N. Sinclair, C. Reimer, M. Zhang, and M. Lončar, "Integrated photonics on thin-film lithium niobate," *Adv. Opt. Photon.* **13**(2), 242–352 (2021).
35. Y. X. Lin, M. Younesi, H. P. Chung, H. K. Chiu, R. Geiss, Q. H. Tseng, F. Setzpfandt, T. Pertsch, and Y. H. Chen, "Ultra-compact, broadband adiabatic passage optical couplers in thin-film lithium niobate on insulator waveguides," *Opt. Express* **29**(17), 27362–27372 (2021).